



Started on	Tuesday, 15 September 2026, 9:16 AM
State	Finished
Completed on	Tuesday, 15 September 2026, 9:19 AM
Time taken	2 mins 21 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int findFirstZero(int arr[], int low, int high) {
4     if (high >= low) {
5         int mid = low + (high - low) / 2;
6         if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0)
7             return mid;
8         if (arr[mid] == 1)
9             return findFirstZero(arr, mid + 1, high);
10        return findFirstZero(arr, low, mid - 1);
11    }
12    return -1;
13 }
14
15 int main() {
16     int m;
17     if (scanf("%d", &m) != 1) return 0;
18     int arr[m];
19     for (int i = 0; i < m; i++)
20         scanf("%d", &arr[i]);
21
22     int idx = findFirstZero(arr, 0, m - 1);
23     printf("%d\n", idx == -1 ? 0 : m - idx);
24     return 0;
25 }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓
✓	10 1 1 1 1 1 1 1 1 1 1	0	0	✓

	Input	Expected	Got	
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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